Snapshot of October 29, 2026. This dossier describes the four repositories as they stood that day; the projects have moved on since, and the repositories are the authority. Underlined links lead to the sources, frozen at their state of that day. What came next lives on gnu6dotlive.github.io.

DOSSIER 3 · THE COMPANION

Calix

A single window to install, update and inventory the software of an Arch Linux system — without ever hiding where each package comes from.

PRE-1.0 | V0.3.1 | GPL-3.0 LICENCE | ARCH LINUX X86_64

6
package ecosystems brought together in a single list

1,803
installed packages in a single table, on the author's machine (October 29, 2026)

4
separate programs: interface, engine, inventory service, notifications

0
calls to `sudo`: elevation goes through a single polkit path

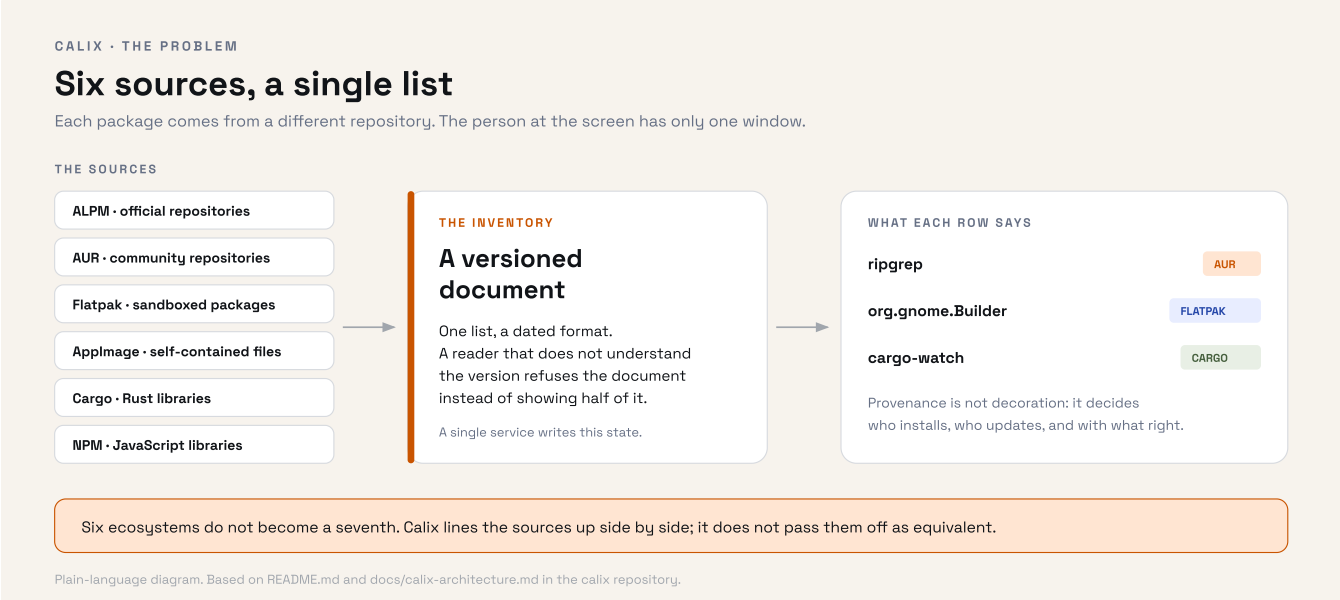
In one picture

THE ANALOGY

Imagine a single counter that accepts parcels from six different carriers. You now have one desk, one tracking number, one queue.

But every parcel keeps its original label: this one came by post, that one by private courier. The counter does not pretend all carriers are the same — it just saves you from running between six desks.

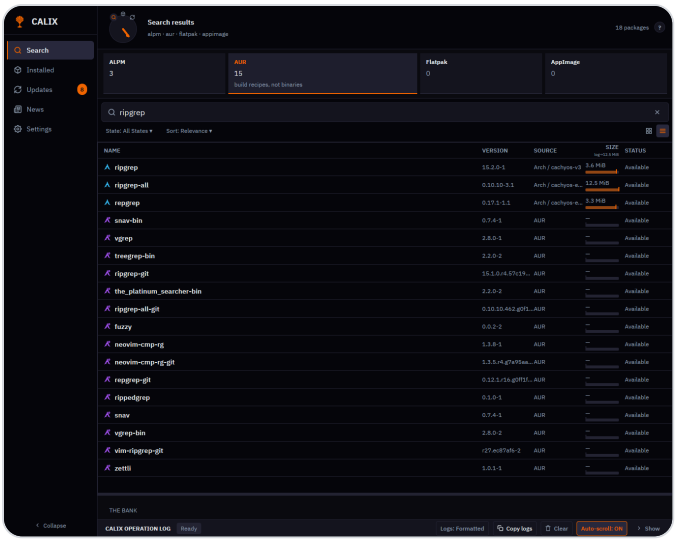
The problem, then the answer

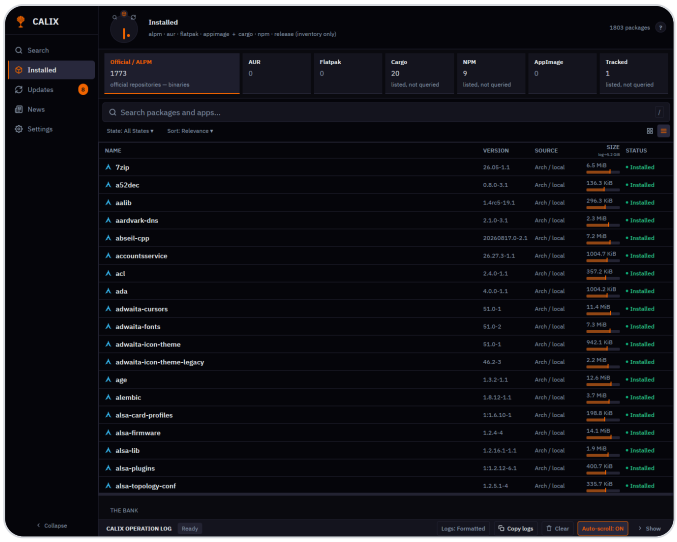


Tidying is not mixing up. Calix brings six ecosystems together in one document, but every entry says where it comes from. A reader that does not understand the document's version refuses it, instead of showing half of it.

On Arch Linux, installing software means moving from one tool to another: `pacman` for the official repositories, a helper for the AUR, `flatpak` for sandboxed packages, and "whatever is lying around" for AppImages. Each has its own output, its own idea of what is installed, and its own path to administrator rights.

Calix brings all of that into a single native application: no embedded web page, no hidden browser. The interface is written in Rust with GPUI — the same foundation as gnosix — and package management in Zig.

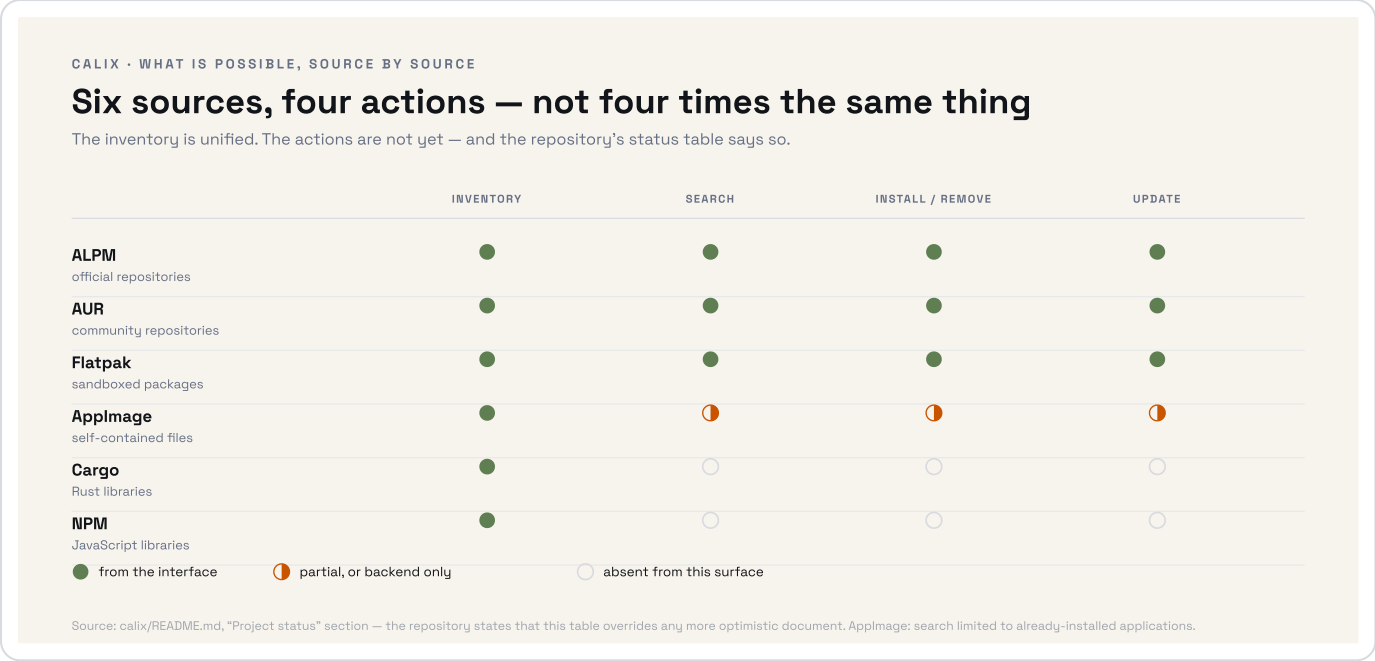




Two real views of Calix. The first, a search, mixes Arch repositories and the AUR: every row keeps its source, and the AUR is flagged as "build recipes, not binaries". The second shows the inventory of the author's 1,803 packages, one tab per ecosystem; Cargo and NPM are "listed, not queried". Screenshots of October 29, 2026, installed version r5022.

What works, what is still open

AREA	STATE PUBLISHED BY THE REPOSITORY	WHAT IT DOES NOT SAY
Interface	Primary interface, in active development	Not a frozen design; the layout can still change.
Unified inventory	ALPM, AUR, Flatpak, AppImage, Cargo, NPM and tracked releases	Listing is not knowing how to install: see the next row.
Actions from the interface	ALPM, AUR and Flatpak only	Cargo and NPM are listed, not yet modifiable from the interface.
Search	ALPM, AUR, Flatpak and already-installed AppImages	The other sources remain inventory-only.
Inventory service	Answers on demand, sole writer of the state	No scheduler yet: nothing updates on its own.
Release tracking	Registry and provenance checks in Zig	A frozen Python engine still writes the observation file.
Notifications	Shipped and verified on a real session	Installed, but not enabled by default.
Platform	Arch Linux, x86_64	No other distribution is supported.



An inventory, not yet a full counter. All six sources are inventoried; only three can be changed from the window. AppImage is handled by the engine, not yet by the interface.

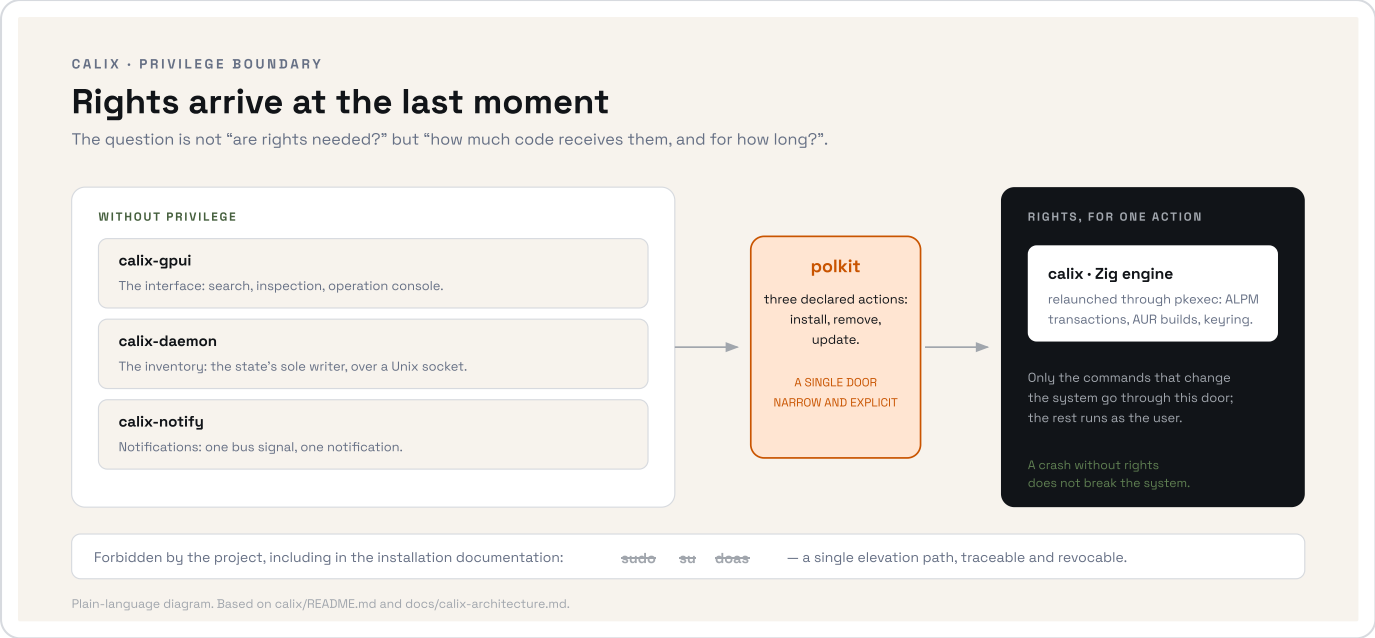
THE REPOSITORY'S RULE

The repository publishes a status table and states that, if another document in the tree contradicts it more optimistically, **this table is the one to trust**. It is a rare discipline, and it is what makes the reading useful.

The privilege boundary

This is the most interesting point for anyone funding or evaluating the project. A package manager needs administrator rights: it changes the system. The question is not "are those rights needed?" but "how much code receives them?".

In Calix, the interface, the inventory service and the notification bridge run **without privilege**. Elevation goes through **a single narrow, explicit path, via polkit**. The project explicitly forbids `sudo`, `su` and `doas` — including in its installation documentation.



A single door. The three everyday programs run under the user’s account. Only the commands that change the system relaunch the Zig engine through `pkexec`, for one of the three declared polkit actions: install, remove, update.

WHAT IT CHANGES

A program that crashes without privilege does not break the system. A program that asks for rights at the last moment, for one precise operation, shrinks the surface of an incident. It is an architectural decision, not a decorative precaution.

Why provenance is the real subject

The common thread of these four repositories is **topology**: making explicit where each thing is, and what that position allows. Calix offers a very concrete version of it.

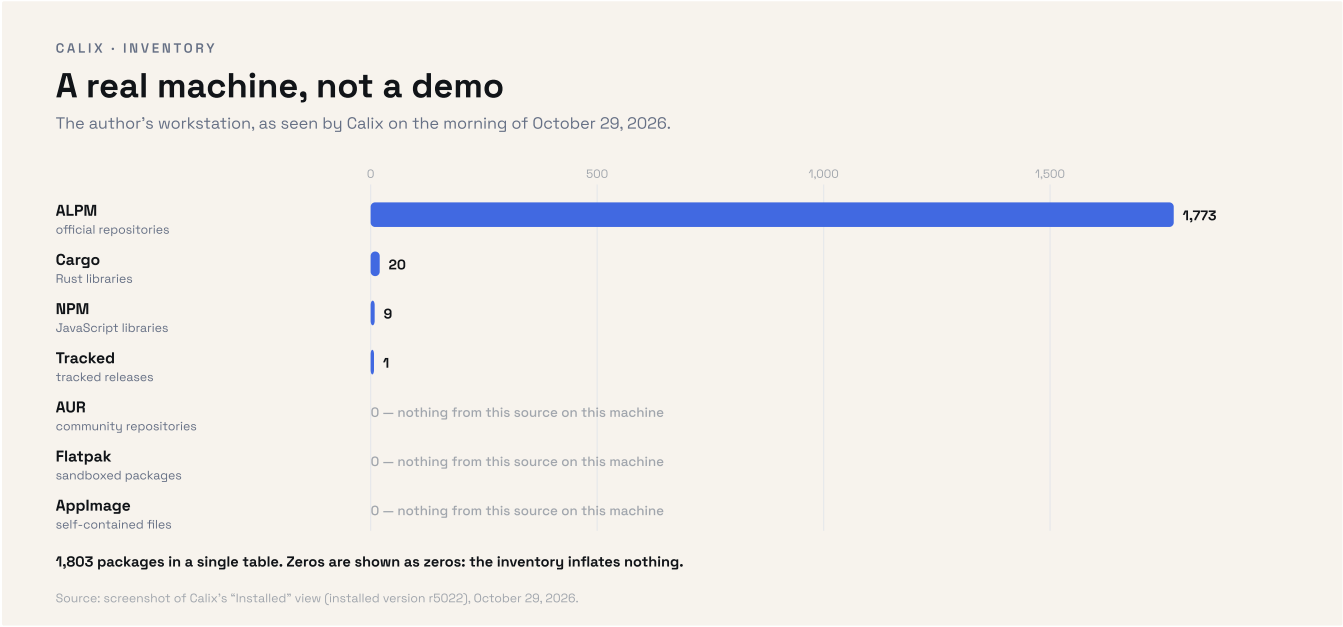
A package installed from the official repositories and the same name coming from the AUR are not the same object. One is signed by the distribution, the other by a community maintainer. Mixing them in an unlabelled list would be a mistake, not a simplification. So Calix keeps the label, and even versions the document format: a consumer that does not know the version refuses to read it.

THE ANALOGY

A parcel label does not tell you whether the contents are good. It tells you who sent it, and therefore whom you will be dealing with if something goes wrong. That is exactly what Calix does with a package.

A real machine, not a demo

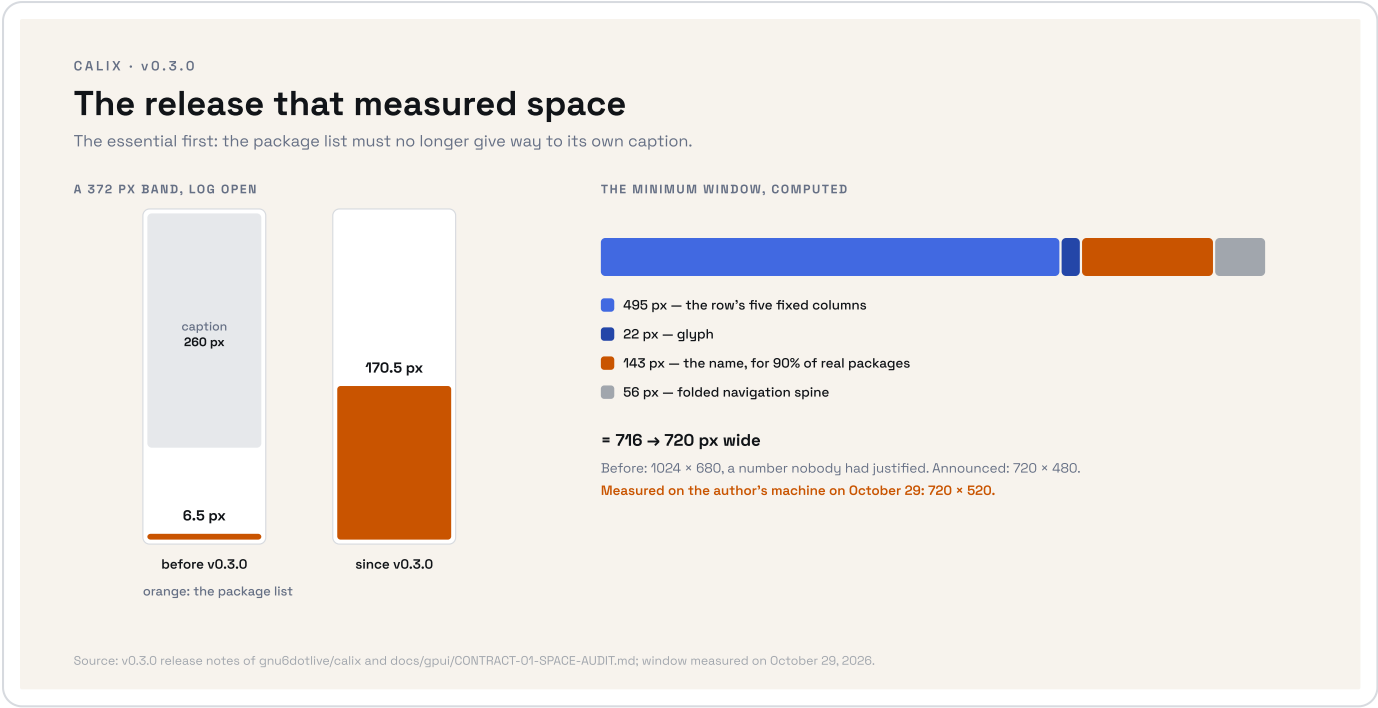
The chart comes from a screenshot of Calix on the author's workstation: 1,803 packages, almost all from the official repositories. This is what an ordinary Arch Linux system looks like, seen all at once.



Zeros are information. No AUR, Flatpak or AppImage in the inventory that day; Cargo and NPM are listed without being queried. An honest inventory does not invent diversity.

History: the release that measured space

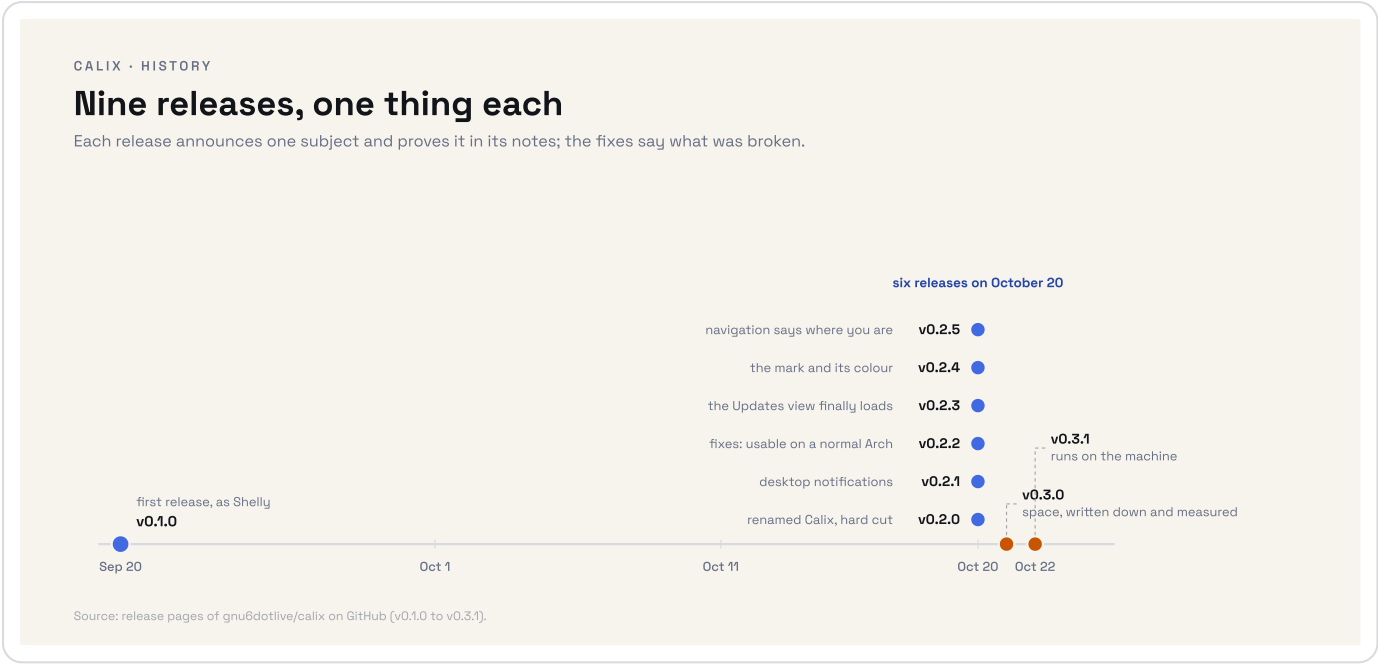
Before v0.3.0, with the log open, the package list shrank to 6.5 pixels high while its own caption kept 260. v0.3.0 wrote down who is entitled to space — and measured it: 170.5 pixels in the same band.



The essential first. Three layers in order: navigation and actions, then the package list, then the context surfaces. Before this release the order was inverted, and that was measurable.

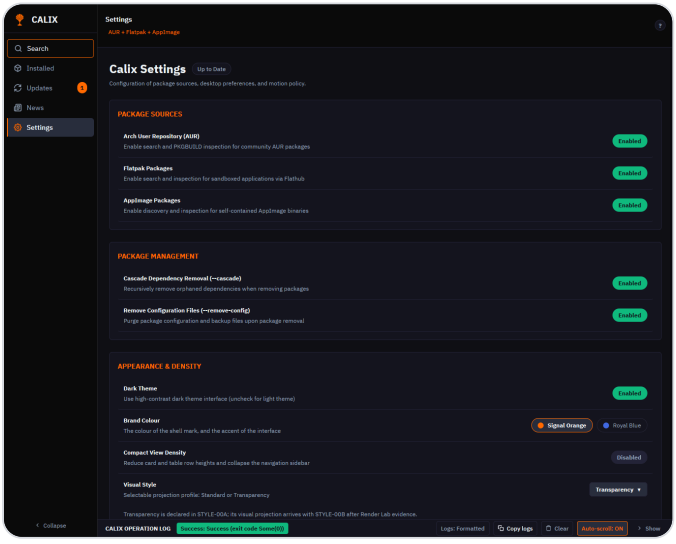
The same release replaced a minimum window size nobody had justified (1024 × 680) with a **computed** one: a row's five fixed columns, the glyph, room for the name of 90% of a real machine's packages and the folded navigation bar add up to 716 pixels, rounded to 720. The case that failed at 560 pixels is kept in the tests. Measured on the author's machine on October 29, the minimum window is 720 × 520; the release note announces 480 high.

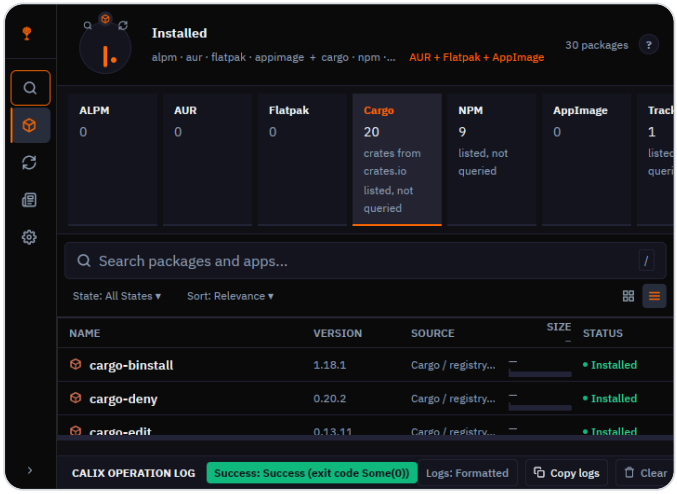
Nine releases, one thing each



Six releases in one day. On October 20 the project changes its name (from Shelly to Calix, with no compatibility layer), then ships notifications, two rounds of fixes, the mark and the navigation. Each note says what was broken before.

The latest, v0.3.1, is "the release that runs on the machine": the interface moves to the current generation of its graphics library — twenty-four compile errors brought down to zero, and **no layout measurement moved**. Where the documents and the code disagreed, the documents were corrected.





Two more views. The settings, with the brand colour to choose, orange or blue (v0.2.4); the window at its minimum size, where the navigation bar folds to make room for the list (v0.3.0). This second screenshot was taken after updates, in a state where the official inventory was not shown: it shows the layout, not the figures. Screenshots of October 29, 2026.

Projections: an even narrower door

The most telling work item concerns privilege: a dedicated root helper and finer polkit actions, so the interface asks for exactly the operation it needs — and nothing more.

The repository ranks five work items by value, with no dates:

1. **Finish requalifying the navigation**: the clicks the synthetic pointer could not exercise, and the focus that should follow a change of place.
2. **Port the rest to Zig**: four release-tracking commands are still in Python.
3. **Wire the application flow**: a dedicated root helper and narrower polkit actions — the door above.
4. **Close the mutation gap**: install and remove Cargo and NPM packages from the interface, search the sources that are inventory-only today.
5. **Cut the frozen Python engine**, once release tracking is fully native.

What it is not

THE LIMITS, PLAINLY

- **Not finished.** The project is explicitly pre-1.0; version 0.3.1 is a step, not an end point.
- **Not universal.** Arch Linux and x86_64 only. It is not a package manager for Debian, Fedora or macOS.
- **Not automatic yet.** The service answers on demand: there is no scheduler, so no background updates.
- **Not everything is modifiable.** Cargo and NPM appear in the inventory but cannot yet be installed from the interface.
- **Not fully native.** A frozen Python engine still writes the tracked-releases file; replacing it is one of the announced work items.
- **Gestures not requalified.** Requalifying the navigation remains open: the clicks could not be exercised by the synthetic pointer.

What it changes, depending on who you are

Arch Linux user

One window instead of four tools, and a list that does not lie to you about where a package comes from. Updates remain manual for now.

Developer

A complete example of a native GPUI interface, a Zig backend talking to `libalpm`, a versioned JSON-lines protocol over a Unix socket, and a polkit boundary written down in black and white.

Decision-maker or funder

The value is not "one more app store": it is a model of provenance and privilege, reusable wherever several sources must coexist without being confused.

Going further

> TECHNICAL — THE REPOSITORY'S WORDS

> CHECKING FOR YOURSELF